

Oriel summer school

Adverse selection and moral hazard

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Outline

1. Adverse selection
2. Akerlof lemons
3. Moral Hazard
4. Incentive pay

Introduction

- Adverse selection arises from **asymmetric information** in economic transactions.
- The party who does not know is aware that he does not know and the other party knows
- This leads to a selection effect where only the people who know engage in the market
- If extreme enough, this selection effect leads to a market break-down

Examples of Adverse Selection

Used car market:

- Seller has more information about the car's quality.
- Buyers know that the sellers know more and are only willing to pay for the average quality car
 - This drives away sellers of good quality cars which worsens the average quality of market cars
 - Buyers have even less willingness to pay
 - Market breakdown

Examples of Adverse Selection

Health insurance:

- Individuals with higher health risks are more likely to purchase insurance
- Insurance companies do not perfectly observe health risks. But they know they will attract less healthy people
 - Insurance companies raise premia, leading to healthier individuals dropping out
 - Insurance companies raise premia even more
 - Market breakdown

Mitigating Adverse Selection

- **Screening and signaling mechanisms:**
 - Used car market: Sellers provide warranties or certified pre-owned programs to signal quality
 - Health insurance: Risk-based pricing, medical exams to assess risk accurately
- **Information disclosure:**
 - Insurance companies request detailed health information ⇒ Concerns about privacy and fairness
 - Online reviews for car dealers
- **Public policy interventions:**
 - Setting up disclosure requirements
 - Mandating insurance to force the good types in

Implications of Adverse Selection

- Adverse selection leads to **suboptimal outcomes**. It can either lead to the over-provision or under-provision of a good
- Markets with gains from trade fail to emerge because of adverse selection
- It affects most sectors of the economy e.g. including insurance, credit markets, and labor markets
- Understanding adverse selection is crucial for **policy design** and market analysis.

Conclusion

- Adverse selection results from **information asymmetry** in economic transactions.
- It can be mitigated through **screening mechanisms, information disclosure** and **public policy interventions**.
- Adverse selection has significant implications for market efficiency and policy design.

Akerlof lemons

- Nobel prize-winning paper that first describes the adverse selection
- Uses the example of the second-hand car markets where you can buy either good cars or bad cars (famously called lemons)
- The key mechanism is that at any given price point, might keep the high-quality cars and sell the lemons. And buyers **know** that

The set up

Assumptions

1. Cars differ in quality, q distributed uniformly on $[0, 1]$
2. M sellers: know quality; each can sell one car; utility of owning a car $U_s = T \cdot q$
 - Sell if $T \cdot q \leq P$
3. N buyers: do not know quality; each wants to buy one car; utility of owning a car $U_b = t \cdot q$
 - Buy if $t \cdot q \geq P$
4. There are potential gains from trade: $t > T$ (t and T are how much the buyers and seller value the car)

The set up (Cont)

Market outcome determined but supply, demand and average quality

Supply: a function of P . Sell the car if $P \geq q \cdot T$, or quality $q \leq \frac{P}{T}$

$$S(P) = \begin{cases} \frac{P}{T} & \text{if } P < T \\ 1 & \text{if } P \geq T \end{cases}$$

Demand: a function of P and average expected quality qa

$$D(P; qa) = \begin{cases} N & \text{if } P \leq t \cdot qa \\ 0 & \text{if } P > t \cdot qa \end{cases}$$

Average quality depends on the supply of cars

$$qa(P) = \begin{cases} \frac{P}{2T} & \text{if } P < T \\ \frac{1}{2} & \text{if } P \geq T \end{cases}$$

Case 1

Suppose that $t = 3/2$ and $T = 1$. What happens to price and allocation?

- Assuming all cars are present in the market consumers are willing to pay $qa \cdot t = \frac{3}{4}$

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- Assuming all cars are present in the market consumers are willing to pay $qa \cdot t = \frac{3}{4}$
 - With a price of $P = \frac{3}{2}$ only seller with quality $q < P = \frac{3}{2}$ are willing to sell.
 - Which means that average quality is now $qa = \frac{3}{4}$ and willingness to pay for buyers is $\frac{3}{8}$
 - This lower willingness to pay further lowers the average quality of the cars and so on ... Up until the point where cars of quality 0 that sell for a price 0 are left!
 - Market breakdown

Case 2

Now suppose that $t = 2$ and $T = 1$. What happens to price and allocation?

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- Assuming all cars are present in the market consumers are willing to pay $qa \cdot t = 1$
 - With a price of $P = 1$ all sellers are willing to sell
 - Which means that the average quality stays fix
 - We now have an efficient allocation where all potential gains from trades are realised and $P = 1$

Conclusion

Conclusion: Efficiency realised only for large enough gains from trade

1. **Case (1):** No trade although there are gains from trade as $t > T$
2. **Case (2):** All cars are sold but winners and losers compare to perfect information. High-quality car sellers lose out and low-quality car sellers gain. Cross-subsidisation like in a pooling equilibrium

Introduction

“Any situation in which one person decides how much risk to take, while someone else bears the cost if things go badly.”

- Paul Krugman

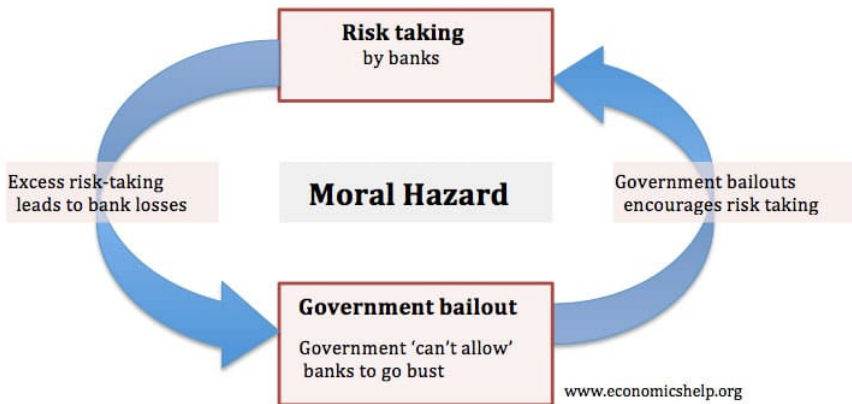
- Moral hazard is a concept in economics that describes a situation where one party's behaviour changes due to protection from the consequences of their actions
- When one does not internalise the consequences of one's actions, market allocation is often suboptimal

Examples of Moral Hazard

Perhaps this will give you some ideas for your end-of-the-week presentation!

- **Comprehensive insurance policies** decrease the incentive to take care of possessions.
- **Government bailouts** of loss-making banks can encourage banks to take greater risks.

Examples of Moral Hazard



Other Examples of Moral Hazard

- **Insurance and consumer behaviour** : When insured, people may have less incentive to take precautions (e.g., bike theft).
- **Sub-Prime Mortgages**: Lenders had less incentive to check mortgages when they could sell them on with all the risk borne by others.
- **Fiscal and Monetary Union**: Membership in a monetary union may lead to excessive debt due to the assumption of bailouts by other countries.
- **Management**: Guaranteed job security may alter incentives and lead to self-serving decisions.
- **Health insurance**: Doctors may over-prescribe treatment since the cost is borne by insurance companies.
- **Moral Hazard from IMF intervention**: Intervention may encourage risky behaviour by countries.

Implications and Mitigation

- Moral hazard leads to distortions and inefficiencies in various contexts.
- Strategies to mitigate moral hazard:
 - Designing appropriate **incentive structures** e.g. bundling health insurance with gym membership
 - Implementing effective **monitoring mechanisms** e.g. app to monitor one's driving
 - Correcting inefficiencies *ex post* e.g. Regulations on maximum leverage for Banks

More specific measures

- **Build in incentives:** Design contracts to give individuals an incentive to take precautions (e.g., deductibles in insurance).
- **Penalize bad behaviour :** Bail out institutions but hold individuals responsible for reckless decisions.
- **Split up banks:** Separate risky investments from consumer savings to avoid too-big-to-fail situations.
- **Performance-related pay:** Introduce performance evaluation and no guarantee of lifetime employment to avoid a moral hazard in the labour market.

Assumptions

- Agent
 - can work, $e = 1$, or shirk, $e = 0$
 - gets a wage w
 - Utility $U(w) - e$, where e is the cost of working and $U(\cdot)$ is concave (risk-averse agent)
 - Does not work if utility lower than U
- Principal
 - Does not observe effort (hidden action)
 - Observes a signal: S, F (success or failure) - the signal realization depends on effort
 - Payoff to the principal
 - S with probability p_e
 - F with probability $1 - p_e$
 - $p_1 > p_0$ (success rate depends on effort)
 - Offers a wage to induce effort

High Effort

- Principal wants to induce high effort, $e = 1$, with wages w_S , w_F
- Principal (the firm) gets:
 - $\Pi_1 = p_1(S - w_S) + (1 - p_1)(F - w_F)$

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- Notice that the firm can fulfil the above constraint and get the wages $w_F + w_S$ to be infinitely low, to maximise its profit. But the worker will quit the job if its utility is below U . Thus wages also need to be fulfilled :
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 - $p_1 U(w_S) + (1 - p_1) U(w_F) - 1 \geq U$ (IR), where U is the reservation utility
- Principal (the firm) gets:
 - $\Pi_1 = p_1(S - w_S) + (1 - p_1)(F - w_F)$

Analysis

When the firm maximises its profit while fulfilling the conditions from the previous slide:

1. (IR) must be an equality. Why? Because otherwise, the principal could increase the payoff by reducing the wage.
2. (IC) must be an equality. Why? Because the agent is risk averse and the principal is risk neutral. The agent bears a utility cost when $w_s - w_f$ increases and the principal has to compensate the agent for this cost. The principle wants to spread $w_s - w_f$ as little as possible.

High-effort

- (IR) and (IC) are an equality $\Rightarrow$ 2 equations in two unknowns w_F, w_S .
- Hence, w_F^* and w_S^* are determined by:

$$U(w_F^*) = U - \frac{p_0}{p_1 - p_0}, \quad U(w_S^*) = U + \frac{1 - p_0}{p_1 - p_0}$$

- The principal gets:

$$\Pi_1 = p_1(S - w_S^*) + (1 - p_1)(F - w_F^*)$$

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- Prefer to induce high effort if:

$$\Pi_1 - \Pi_0 > 0$$

or,

$$[p_1 - p_0] \cdot (S - F) + w - p_1 w_S^* - (1 - p_1) w_F^* > 0$$

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- Notice: w_F^* and w_S^* do not depend on S , F .
- Result: If $(S - F)$ is large, then the principal prefers to induce high effort.

Manager Pay and Performance

- Manager pay should depend on manager performance.
- It is increasingly the case in practice. Again at the core of the problem is the asymmetry of information. The effort is private information.
- Note that the term "effort" is not meant literally. For example, it can stand for the worker overusing company resources for his benefit: e.g. the manager using the company's private jet to go on vacation!